

Hydrogen energy; Solar-hydrogen revolution

We are going to learn about hydrogen energy which is another important renewable energy resource.

Let's begin with **what is hydrogen energy?**

Hydrogen is one of the simplest elements consisting of one electron and one proton. This simple element is always found combining with other elements and forms compounds like water H_2O . Moreover, it is most abundantly present in nature. But naturally, hydrogen is not present in its gaseous form on the earth.

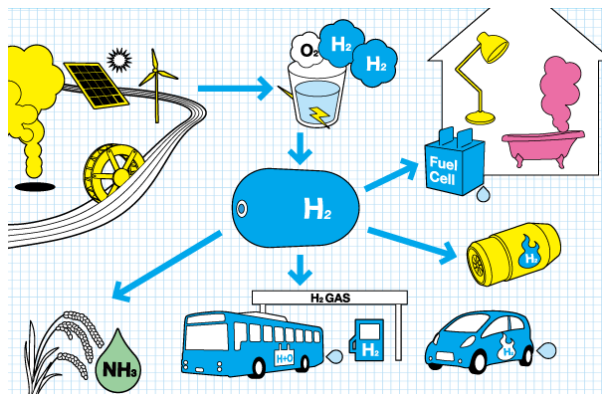


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It is very light energy-dense fuel. It can be stored safely in hydrogen tanks. Burning of Hydrogen fuel produces no greenhouse gas emission such as carbon dioxides and other pollutants. Therefore, it is a clean and environmentally friendly fuel. Hydrogen when combined with oxygen in a fuel cell produces heat and electricity with the liberation of water vapor as a by-product.

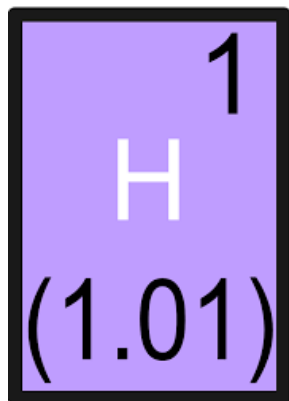


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Hydrogen can be produced from fossil fuels or biomass or it can be produced by passing electricity through water. That is Electrolytic splitting of the water into its constituent components hydrogen and oxygen.

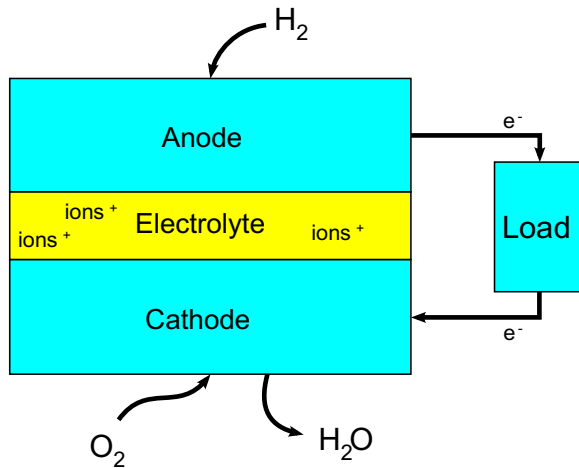


Photo courtesy **wikipedia**

Hydrogen is not an energy source, but it is an **energy carrier**. Because hydrogen does not exist freely in nature, and it is only produced from other sources of energy that's why it is known as an *energy carrier*. This means that hydrogen has to be produced from one of the primary energy sources: such as fossil fuels nuclear energy or renewable energy resources such as solar wind biomass hydro energy or geothermal energy.

As I mention hydrogen does not exist freely in nature then how to produce it?

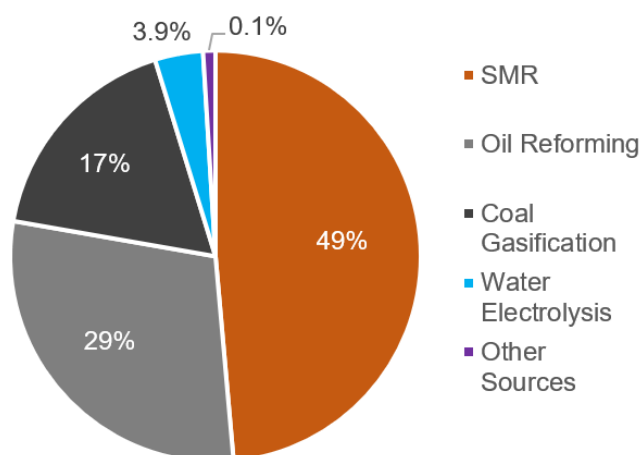
Hydrogen can be produced from a variety of resources such as natural gas nuclear power biomass and renewable power like solar and wind. These qualities make hydrogen as an attractive fuel option for transportation sector and electricity generation applications. It can be used in cars used in houses for portable power generation.

Today hydrogen fuel can be produced through several methods.

The most common methods today are

1. Natural gas reforming this is a thermal process
2. Thermochemical Water Splitting
3. electrolysis.
4. solar-driven processes or solar hydrogen cycle
5. biological processes.

Global Hydrogen Production by Method



(Dincer & Acar, 2015)

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We can discuss above methods one by one

First what is natural gas reforming or thermal processes?

Thermal processes for hydrogen production typically involve steam reforming. That is a high-temperature process in which steam reacts with a fuel which contains hydrocarbons to produce hydrogen. Many hydrocarbon containing fuels can be converted into hydrogen including natural gas diesel renewable liquid fuels coal or biomass. Today >95% of hydrogen is produced from steam reforming of natural gases.

But there is disadvantages associated with this process 1. natural gas is already rapidly becoming limited source and more expensive one. 2. It is also a fossil fuel so the carbon dioxide which is released in the reformation process increase the greenhouse effect and global warming.

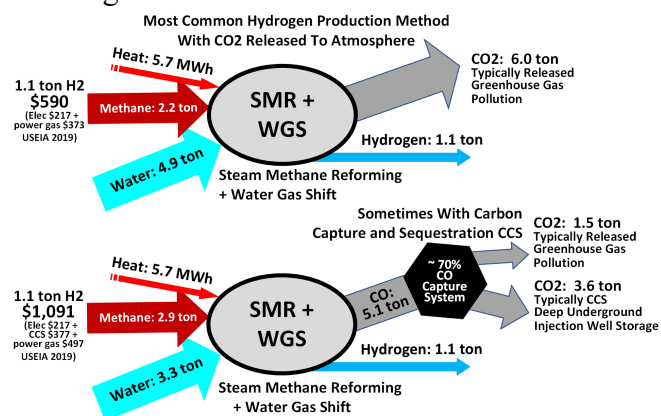


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Let's move to the second method of Hydrogen Production That is Thermochemical Water Splitting

Thermochemical water splitting uses high temperatures nearly 500°–2000°C which may come from concentrated solar power or from the waste heat from the nuclear power stations. In this process chemical reactions would take place and then produce hydrogen and oxygen from water. The chemicals such as cerium oxide or copper chloride can be used in the processes. Usually these chemicals are reused within each cycle and consume only water and produce hydrogen and oxygen.

Solar- or nuclear-driven high-temperature thermochemical water-splitting cycles produce hydrogen with near-zero greenhouse gas emissions using water and sunlight or nuclear energy.

Hydrogen On Demand

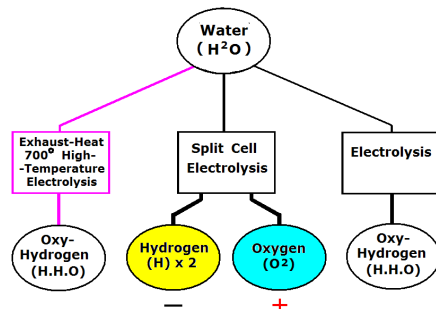


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Let's move to the third method of Hydrogen Production ELECTROLYTIC PROCESSES.

Water can be splitted into oxygen and hydrogen through a process called electrolysis. Electrolysis is the process of using electricity to split water into hydrogen and oxygen. Electrolysis is a promising option for carbon-free hydrogen production from renewable and nuclear resources.

Electrolytic processes take place in an electrolyzer. An electrolyzer creates hydrogen from water molecules. Electrolyzers can range from small-scale hydrogen production to large-scale production facilities that is well-suited for non-greenhouse-gas-emitting electricity production. Electrolyzers consist of an anode and a cathode which is separated by an electrolyte. Different electrolyzers function in different ways mainly due to the different type of electrolyte material involved and the ionic species it conducts.

- First Water reacts at the anode to form oxygen and positively charged hydrogen ions and electrons.
- The electrons flow through an external circuit and the protons selectively moves across the polymer electrolyte membrane (PEM) to the cathode.
- At the cathode hydrogen ions or protons combine with electrons from the external circuit to form hydrogen gas.

Anode Reaction is $2\text{H}_2\text{O}$ reacts and form $\text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

Cathode Reaction is $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$

Hydrogen produced via electrolysis can result in virtually zero greenhouse gas emissions depending on the source of the electricity used. however the production cost needs to be reduced in the future.

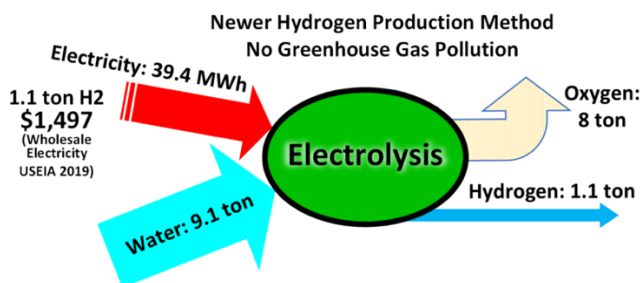


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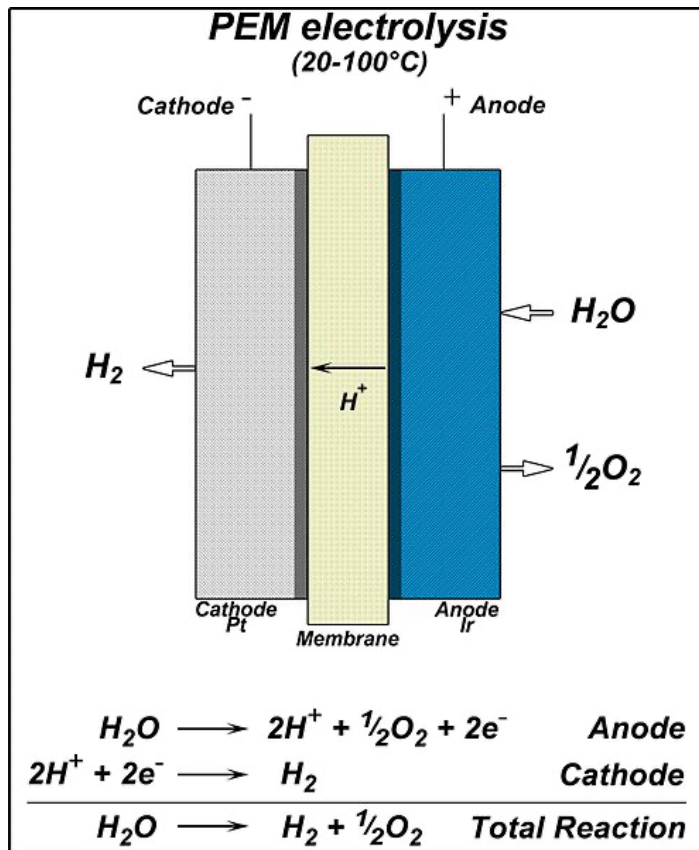


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Let's move to the fourth method of Hydrogen Production SOLAR-DRIVEN PROCESSES

Solar-driven processes use light as the agent for hydrogen production.

A typical Solar-hydrogen energy cycle consist of three components. First one is electrolyzer. Second one is hydrogen storage tank and third one is hydrogen fuel cell.

In a solar-hydrogen energy cycle a solar powered electrolyzer is used to convert water into hydrogen and oxygen. Hydrogen and oxygen produced thus are stored in containers for later use.

The stored hydrogen can be used by a fuel cell to produce electricity when no sunlight is available. Solar-hydrogen energy cycle is also called as Solar-hydrogen revolution

There are a few solar-driven processes including 1. Photobiological processes 2. Photoelectrochemical processes and 3. solar thermochemical processes.

Photobiological hydrogen production processes use the natural photosynthetic activity of bacteria and green algae to produce hydrogen. The photobiological process uses microorganisms and sunlight to turn organic matter into hydrogen.

Photoelectrochemical processes use specialized semiconductors to split water into hydrogen and oxygen.

Solar thermochemical hydrogen production uses concentrated solar power to drive water splitting reactions often with the help of reagents such as metal oxides.

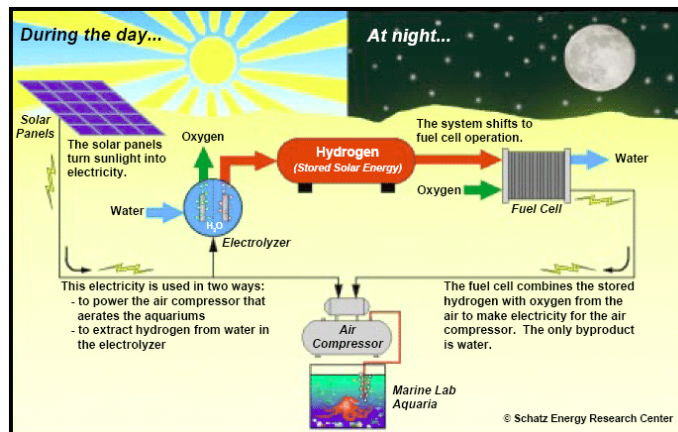


Photo courtesy Ali, Suhaib. (2022). SOLAR-HYDROGEN SYSTEMS FOR REMOTE AREA POWER SUPPLY.

THE LAST METHOD OF HYDROGEN PRODUCTION BIOLOGICAL PROCESSES

Biological processes use microbes such as bacteria and microalgae and can produce hydrogen through biological reactions. In microbial biomass conversion the microbes break down organic matter like biomass or wastewater to produce hydrogen

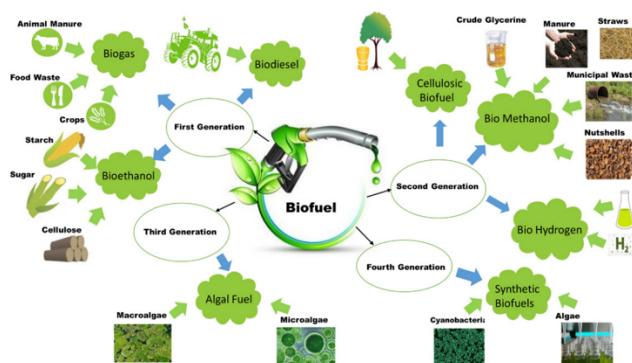


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Let me explain how to utilize hydrogen for generating heat or electricity

Typically Hydrogen Fuel cells are used to generate electricity

Let's discuss what is meant by **Hydrogen Fuel Cell?**

To know about hydrogen fuel cell at first you should know what a fuel cell is?

A fuel cell is a device that can convert the chemical energy into electrical energy.

Thus often this fuel cell is compared to batteries. Both fuel cells and batteries produce energy via chemical reactions and transfer that energy into usable electric power.

In a typical fuel cell hydrogen gas is supplied to anode. At anode these hydrogen molecules divide into protons and electrons. The produced electrons flow through wires and generate electricity while the protons pass through a membrane and combine with oxygen gas at cathode to form water vapor. Overall Hydrogen fuel cells produce electricity by combining hydrogen and oxygen atoms. Due to the applications of hydrogen fuel cells these two elements react and produce a huge amount of energy.

In fuel cells three product we can harvest heat electricity and water.

Different types of fuel cells are available for a wide range of applications.

Small fuel cells can power laptop and computers and even cell phones. Large fuel cells can supply electricity to electric power grids and supply emergency power in buildings and supply electricity in places that are not connected to electric power grids.

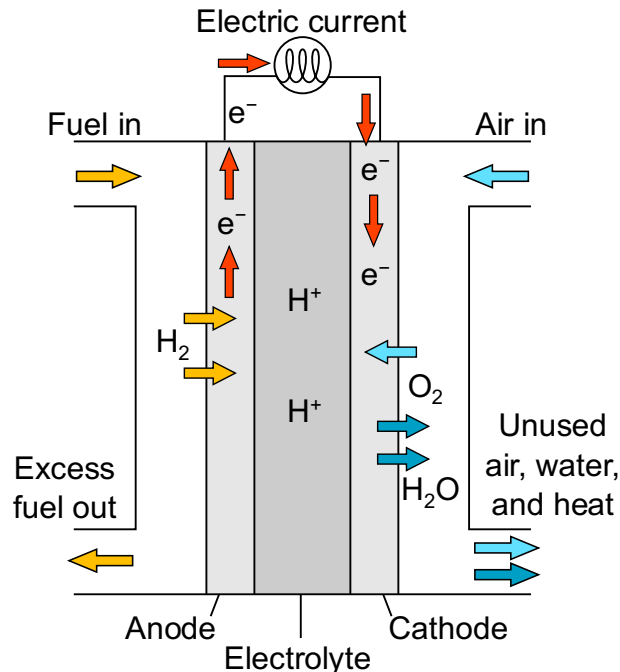


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In this module we should learn another important terminology that is Hydrogen Economy

The term *“hydrogen economy”* refers to the vision of using hydrogen as a low or zero carbon energy sources. For example replacing natural gas as a heating fuel or replacing gasoline as a transport fuel.

It is expected that in a near future hydrogen is produced from a variety of energy sources stored for later use transported to where it is needed and then converted into heat and electricity.

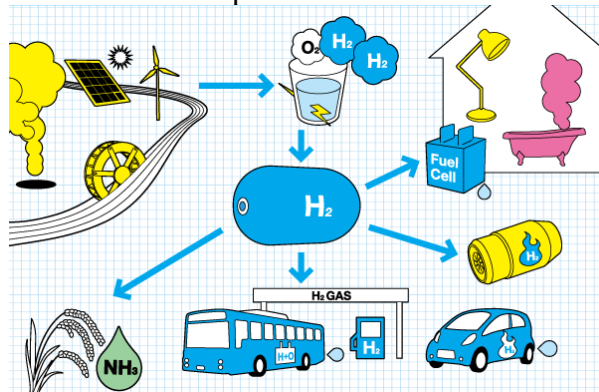


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Let me explain various applications of hydrogen fuel cells

hydrogen fuel cells would eliminate most of the air pollution problems we face today.

It would also greatly reduce the threats of global warming and climate change because using it emits no carbon dioxides.

Hydrogen also provides more energy per gram than does any other fuel making hydrogen the ideal aviation fuel.

Hydrogen is considered as an alternative vehicle fuel. A fuel cell may be two to three times more efficient than an internal combustion engine running on gasoline.

Today hydrogen is used mostly in oil refining industries and fertilisers industries.

Another promising application is in homes where a fuel-cell stack, about the size of a refrigerator could provide heat hot water and electricity.

Honda company has developed a home unit hydrogen generator that produces hydrogen from the methane.

Many Japanese homeowners using such units to produce their electricity and hot water

Finally we can discuss now Advantages and disadvantages of hydrogen energy

Advantages: Hydrogen can be produced from ample water. It has low environmental impact. It is considered as renewable energy if hydrogen is produced from the renewable energy for example as indicated in solar hydrogen cycle. It is a good substitute for oil. It is easier to store than the electricity and safer than gasoline and natural gas. Today many high efficiency hydrogen fuel cells are commercially available and are in developmental stages.

Disadvantages: Hydrogen is not found in nature as H_2 molecule. hydrogen is chemically locked-up in water and in organic molecules such as methane and gasoline. Hence we need to spend an energy to produce hydrogen from these compounds. Therefore net energy is negative. It is difficult to store hydrogen in fuel tank which is the main problem of using hydrogen fuel cell in cars.

Fuel cells are the best way to use hydrogen to produce electricity to operate vehicles but current versions of hydrogen fuel cells are expensive.